Selection on Observables III: Propensity Scores, Weighting, Regression

PS200C, Spring 2025

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Roadmap

1. Theory: Potential outcomes, identification, key quantities
2. Randomization
   - difference in means, variance estimation
   - covariate adjustment
   - blocking
3. Selection on Observables
   - sub-classification
   - matching (on X)
   - weighting (on X)
   - matching and weighting on $Pr(D = 1)$ (propensity scores)
   - regression
4. Instrumental Variables
5. Regression Discontinuity
6. Difference in Differences and Synthetic Control
7. Sensitivity and Bounds

We have to keep saying this...

Assumption: Conditional Ignorability

$$\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp D_i \mid X_i = x \quad \text{for any} \quad x \in \mathcal{X}$$

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- This will allow us to estimate things as if we have done so conditionally on X then averaged over some density in X

Propensity Scores: The conventional motivation

- May be tough to get close matches on X_i when multi-dimensional
- What if you could just match on a one-dimensional summary?

Conditioning on the Propensity Score

Under SOO and common support,

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The intuition:

- Among those with $\pi(X_i) = c$, you have $Pr(D_i = 1) = c$ no matter what differences in X_i you see.
- That is, D_i does not depend upon X_i, conditionally on $\pi(X_i)$.
- Once D_i no longer depends on X_i, it also no longer depends on $\{Y_{1i}, Y_{0i}\}$ (SOO).
- So its like random experiments within each stratum of $\pi(X_i)$.

Identification with Propensity Scores

Proof.

Show that $\Pr(D = 1 | Y_0, Y_1, \pi(X)) = \Pr(D = 1 | \pi(X)) = \pi(X)$, implying independence of (Y_0, Y_1) and D conditional on $\pi(X)$.

$$\Pr(D = 1 | Y_1, Y_0, \pi(X)) \quad = \quad \mathbb{E}[D | Y_1, Y_0, \pi(X)]$$

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therefore $\Pr(D = 1 | Y_1, Y_0, \pi(X)) = \Pr(D = 1 | \pi(X))$ □

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  - But if using weights, check how extreme they get and how few units are doing most of the work

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IPW weights that also guarantee balance?

- Covariate balancing propensity scores (CBPS; Imai, Ratkovic 2014)
- For continuous treatments, CBGPS (Fong, Hazlett, Imai 2017)

Using Propensity Scores: Blattman & Annan

```
pi.out = glm(abd~age+fthr_ed+mthr_ed+hh_size96+orphan96+C_ach+C_akw+
   C_ata+C_kma+C_oro+C_pad+C_paj+C_pal,data=dat,family="binomial"(link=log

matchout.pi=Match(Y=dat$educ,Tr=dat$abd,X=pi.out$fit,estimand="ATT")
summary(matchout.pi)

#Check balance
mb.out.pi =MatchBalance(match.out=matchout.pi,abd~age+fthr_ed+
   mthr_ed+hh_size96+orphan96+C_ach+C_akw+C_ata+C_kma+
   C_oro+C_pad+C_paj+C_pal,data=dat)

btest_after_pi=baltest.collect(mb.out.pi,var.names=varnames,after=TRUE)

balancecompare.pi=cbind(btest[,c("mean.Tr","mean.Co","T pval",
"KS pval")],btest_after_pi[,c("mean.Tr","mean.Co","T pval","KS pval")])
```

Using Propensity Scores: Blattman & Annan

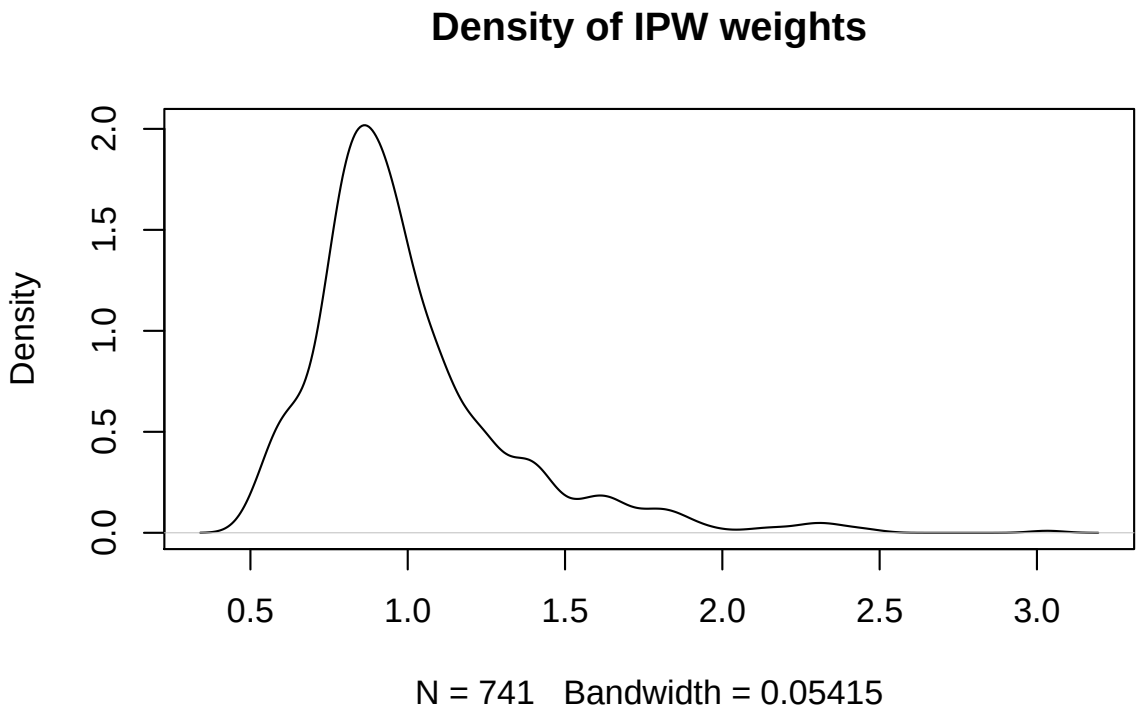
```
 balancecompare.pi
          mean.Tr mean.Co T pval KS pval mean.Tr mean.Co T pval KS pval
age         21.37   20.15   0.00    0.00   21.37   21.04   0.27    0.17
fthr_ed      5.76    6.07   0.27    0.87    5.76    5.60   0.48    0.09
mthr_ed      2.09    2.49   0.09    0.35    2.09    2.40   0.11    0.20
hh_size96    8.09    8.70   0.06    0.04    8.09    8.02   0.79    0.03
orphan96     0.08    0.08   0.90      NA    0.08    0.11   0.07      NA
C_ach        0.15    0.11   0.13      NA    0.15    0.17   0.60      NA
C_akw        0.16    0.08   0.00      NA    0.16    0.17   0.62      NA
C_ata        0.10    0.20   0.00      NA    0.10    0.09   0.55      NA
C_kma        0.15    0.12   0.19      NA    0.15    0.16   0.87      NA
C_oro        0.05    0.14   0.00      NA    0.05    0.05   0.95      NA
C_pad        0.12    0.12   0.98      NA    0.12    0.09   0.10      NA
C_paj        0.15    0.10   0.06      NA    0.15    0.15   0.91      NA
C_pal        0.11    0.13   0.51      NA    0.11    0.13   0.45      NA
```

Using Propensity Scores: Blattman & Annan

Weighting on the propensity scores (stabilized IPW):

```
ps=pi.out$fit
D=dat$abd
PrD=mean(D)
IPW= (D*PrD+(1-D)*(1-PrD))/(D*ps+(1-D)*(1-ps))

#Good to check how crazy the weights are:
plot(density(IPW))
```



How does $p(\pi_i)$ look before/after weighting?

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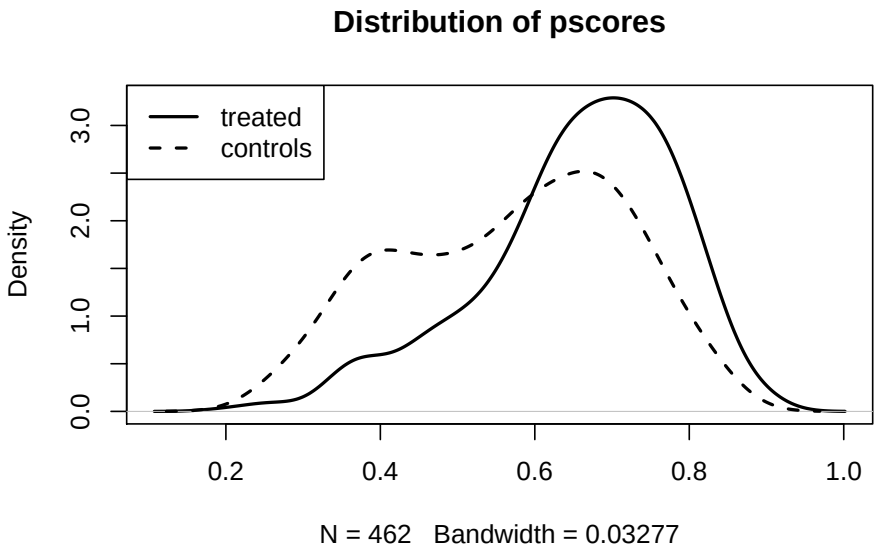
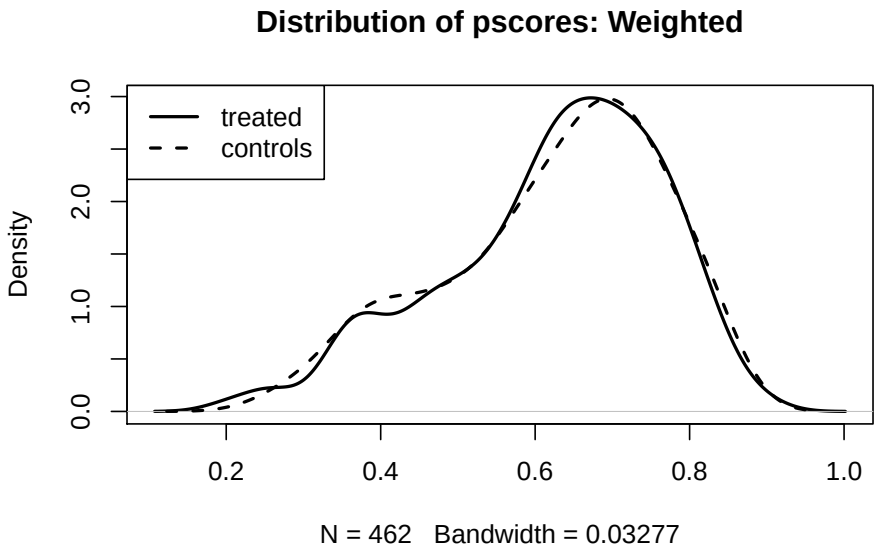
```
plot(density(pi.out$fit[D==1], weight=IPW[D==1]/sum(IPW[D==1])))
lines(density(pi.out$fit[D==0], weight=IPW[D==0]/sum(IPW[D==0])),lwd=2)
legend("topleft", legend=c("treated","controls"), lty=c(1,2))
```





Check Omnibus Balance

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omnibus.bal = lm(abd~age+fthr_ed+mthr_ed+hh_size96+orphan96+
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summary(omnibus.bal)
#R2=0.08, F-statistic: 5.585 on 12 and 728 DF,  p-value: 3.303e-09
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omnibus.bal.ipw = lm(abd~age+fthr_ed+mthr_ed+hh_size96+orphan96+
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Coefficients: (1 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.138e-01  1.058e-01   5.804 9.69e-09 ***
age          5.474e-04  3.691e-03   0.148    0.882
fthr_ed     -4.053e-04  5.352e-03  -0.076    0.940
mthr_ed     -2.116e-04  6.628e-03  -0.032    0.975
hh_size96    4.875e-04  4.689e-03   0.104    0.917
orphan96     3.979e-03  6.873e-02   0.058    0.954
---
Multiple R-squared:  0.0002747, Adjusted R-squared:  -0.0162
F-statistic: 0.01667 on 12 and 728 DF,  p-value: 1
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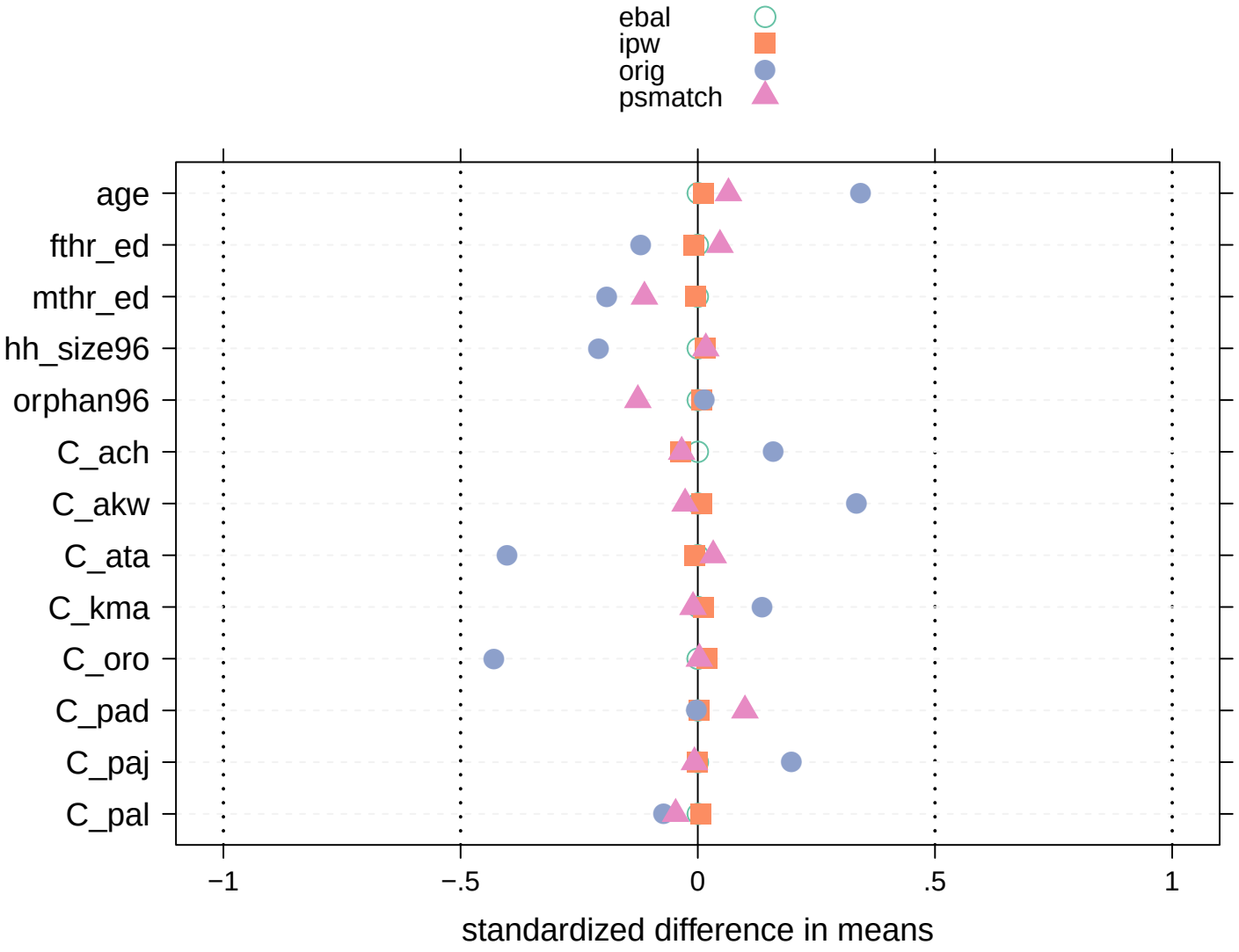
Yay!

Estimating the ATE with IPW weights

```
lm.out.naive = lm(educ~abd, data=dat)
lm.out.ipw = lm(educ~abd, weight=IPW, data=dat)
lm.out.ipw2=lm(educ~abd+age+fthr_ed+mthr_ed+hh_size96+orphan96+
    C_ach+C_akw+C_ata+C_kma+C_oro+C_pad+C_paj+C_pal,weight=IPW,data=dat)
```

	Dependent variable:		
	educ		
	(1)	(2)	(3)
abd	−0.595***	−0.726***	−0.735***
	(0.218)	(0.220)	(0.210)
Constant	7.416***	7.499***	6.113***
	(0.172)	(0.174)	(0.610)
Observations	741	741	741
R^2	0.010	0.014	0.128
Adjusted R^2	0.009	0.013	0.112
Residual Std. Error (df = 739)	2.876	2.904	2.754
Note:		*p<0.1; **p<0.05; ***p<0.01	

Balance Plot after IPW



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Preview: we could get this if we could "move" the treated/control to both have $p(X)$ instead of $p(X|D = 1)$ and $p(X|D = 0)$

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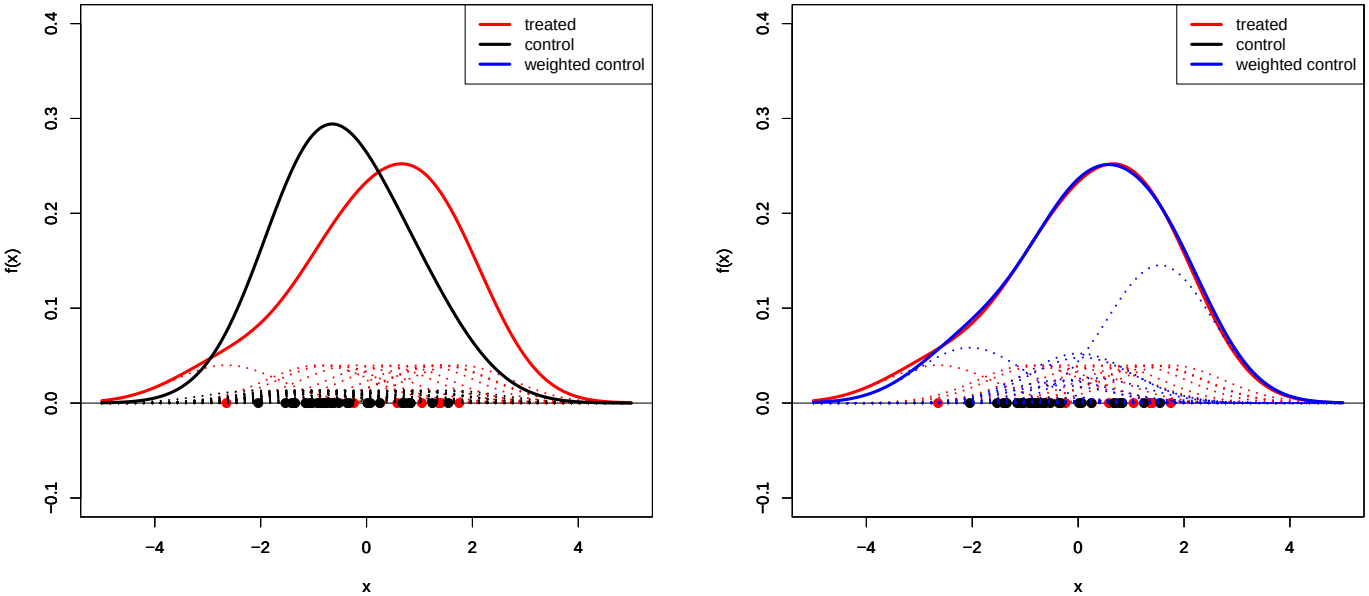

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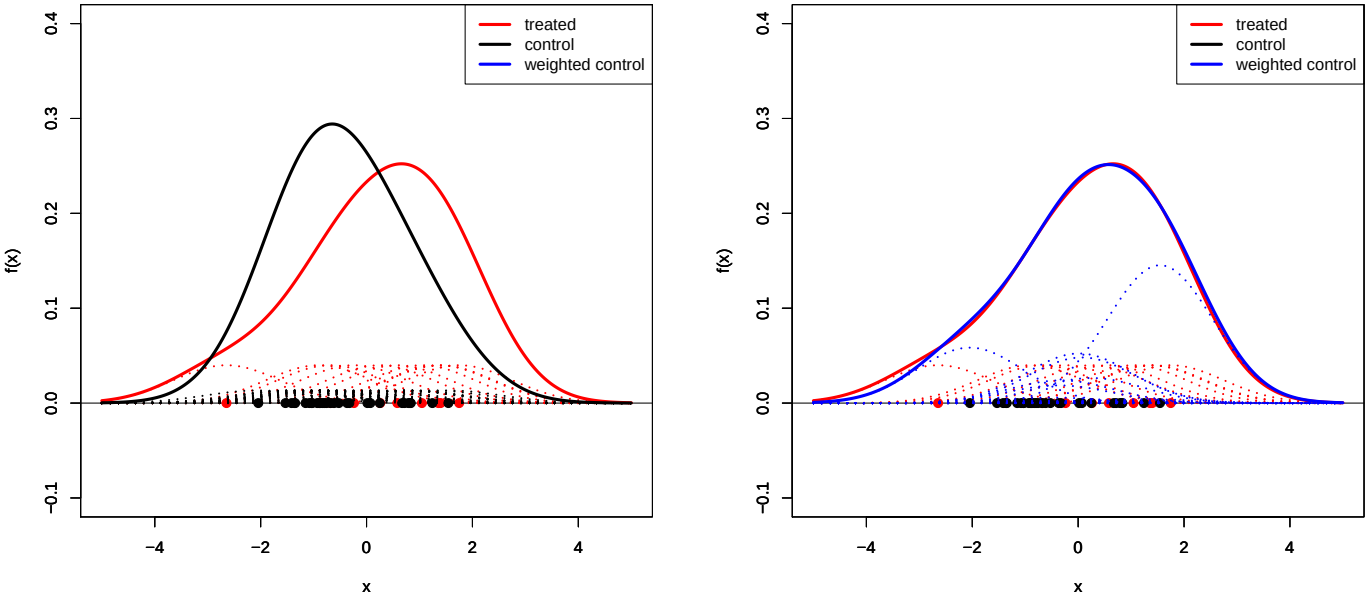

Here we move a control distribution to look like treated (as you would for ATT), but the point is about the role of weights in shifting a distribution.

What weights achieve this?

For ATE, we wish we could average over $p(X)$ for treated and controls,

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$$\begin{aligned}
\mathbb{E}[\hat{\tau}_{IPW}] &= \int \mathbb{E}[Y_i|D_i = 1, X_i]\frac{p(X|D = 1)p(X_i)}{p(X_i|D_i = 1)}dx \\
&\quad - \int \mathbb{E}[Y_i|D_i = 0, X_i]\frac{p(X|D_i = 0)p(X_i)}{p(X_i|D_i = 0)}dx \\
&= \int \mathbb{E}[Y_i|D_i = 1, X_i]p(X_i)dx - \int \mathbb{E}[Y_i|D_i = 0, X_i]p(X_i)dx
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- try to compute the IPWs...but this requires a model, $\widehat{p(D_i|X_i)}$
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The first has a longer history and we've talked about it, let's get to the second...

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- It turns out that these get you unbiased ATT estimates when Y_{0i} is linear in the things you get mean balance on

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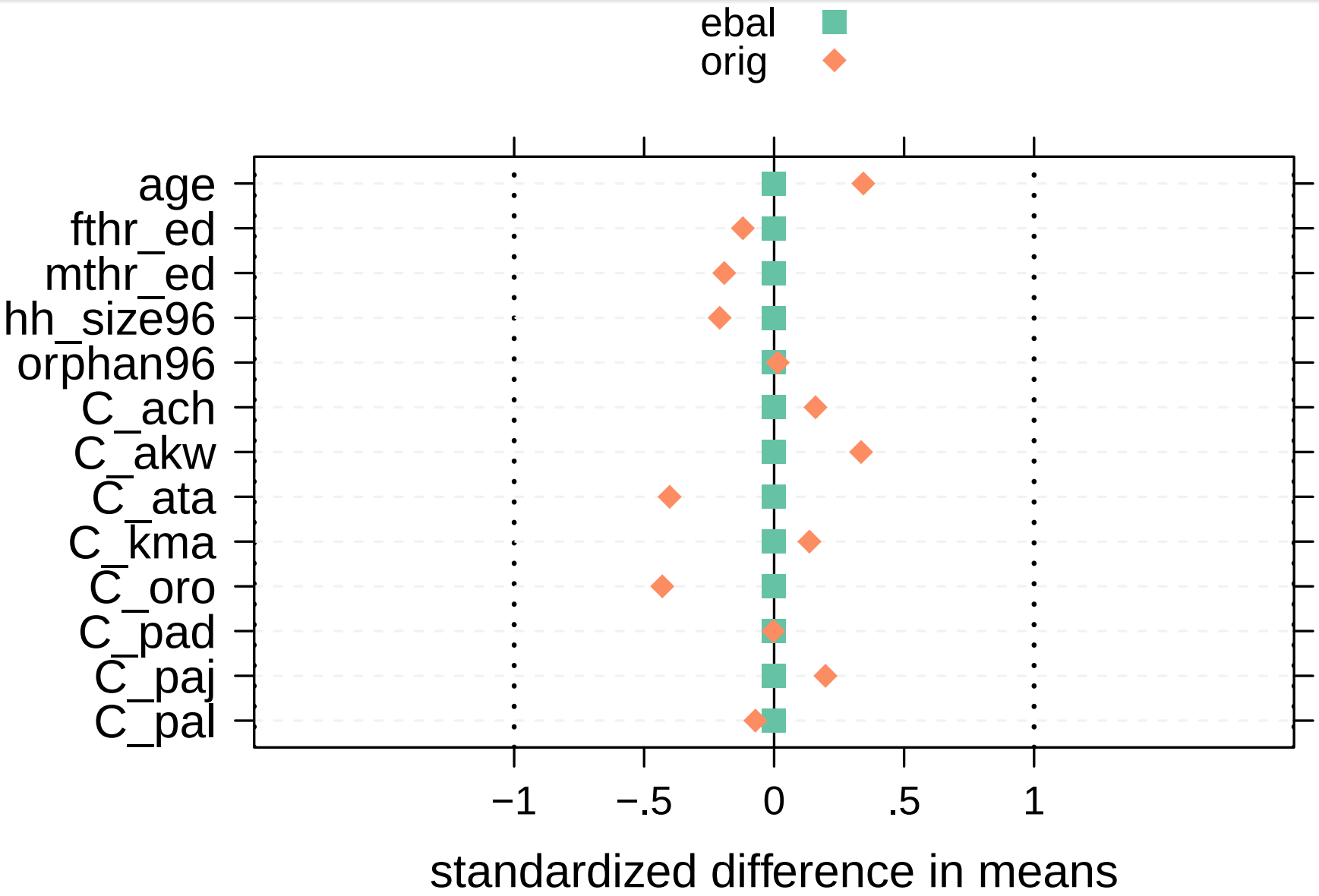
To get multivariate (smoothed) balance, $\hat{p}(X|D=0) = \hat{p}(X|D=1)$, Kernel Balancing (`kbal`, see my website).

Balance on Blattman data with Ebal



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Noting that (2) implies (1) there are 3 possible scenarios:

1. Both (1) and (2) are true.
2. Only (1) is true.
3. Neither (1) nor (2) is true.

Case 1: Constant Effect w/ Linear Potential Outcomes

Result: If treatment effect is constant across units and potential outcomes are linear in X_i, then the OLS estimate of β_1 in the following regression model

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Proof (just how we got here):

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\mathbb{E}[\beta_1] &= \mathbb{E}_x \mathbb{E}[Y_i | D_i = 1, X_i] - \mathbb{E}[Y_i | D_i = 0, X_i] \quad \text{(correct specification)} \\
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Note equivalence: CI+linearity $\rightarrow \varepsilon$ cannot be related to D given X (CIA):

$$\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp D_i \mid X_i \quad \Longrightarrow \quad \varepsilon_i \perp\!\!\!\perp D_i \mid X_i$$

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Result: The OLS estimator can be written as a subclassification estimator, where the weights come from conditional variances of D_i in each subgroup:

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$$\hat{\tau}_{ATE} = \sum_{x \in \mathcal{X}} \{\mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x]\} \Pr(X_i = x)$$

Similarly, the subclassification estimator for the ATT:

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Result: The OLS estimator can be written as a subclassification estimator, where the weights come from conditional variances of D_i in each subgroup:

$$\hat{\beta}_{OLS} = \sum_{x \in \mathcal{X}} \{\mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x]\} \frac{\mathrm{Var}(\hat{D}_i \mid X_i = x)\Pr(X_i = x)}{\sum_{x'} \mathrm{Var}(\hat{D}_i \mid X_i = x') \Pr(X_i = x')}$$

- Intuition: OLS "learns more" from strata of X where π_X closer to 0.50.
- think about what this means for your estimate, especially with lack of overlap
- This is from Angrist and Pischke. It depends on D being linear in X. For more general weights, see my Shinkre & Hazlett.

OLS as a Subclassification Estimator

Estimator	Weights for Subgroups	Unbiased for
$\hat{\tau}_{ATE}$	$\Pr(X_i = x)$	τ_{ATE}
$\hat{\tau}_{ATT}$	$\Pr(X_i = x \mid D_i = 1)$	τ_{ATT}
$\hat{\beta}_{OLS}$	$\dfrac{\mathrm{Var}(\hat{D}_i \mid X_i = x)\Pr(X_i = x)}{\sum_{x'} \mathrm{Var}(\hat{D}_i \mid X_i = x')\Pr(X_i = x')}$	"$\tau_{\text{CVW-ATE}}$"

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- This result assumes discrete Xs, but intuition holds for continuous Xs.

Improvements over pooled OLS

- **Imputation**: use regression to impute missing potential outcomes. E.g. Estimate $\mathbb{E}[Y_{0i}|X_i]$ and use as counterfactuals for observed Y_{1i}. Can weight each pairwise difference as you like. Aka g-computation, T-learner.

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It turns out these are precisely the same thing!

- Both relax your (modeling) assumption from "Y is linear in D and X" (which implies constant treatment effects) to "Y(1) is linear in X, and Y(0) is linear in X", which we call "separate linearity" (Shinkre & Hazlett)
- Unless you somehow care about $\tau_{CVW-ATE}$ as such, the only reason to use OLS without interaction/imputation is that it would be more efficient in the case where you really do have constant treatment effects.

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- "Doubly-robust" estimation by weighted regression:
  - Use weighting (e.g. IPW or balancing weights) in a regression that also includes covariates
- Construct estimators that employ both an outcome model and a weighting of residuals from it.
  - See especially double-ml (DML), augmented IPW (AIPW), targeted maximum likelihood (tMLE)

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- Matching, weighting and regression are used estimate average causal effects when one can assume conditional ignorability
- These are estimation strategies; the validity of the identification strategy (SOO) remains a first-order concern
- No single method is dominant at the moment, and the tides will change...
- My two cents: Ideally the estimator doesn't matter so much; best if you can use 5 different ones and get similar results. Differences in results often point to poor overlap/ model dependency problems that are worth exploring anyway.